Low-Level Design – TFA Settings

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** Low-Level Design · **Version:** 1.0 · **Status:** Draft **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026 **Author:** Jojo George (jojogeorge3344@gmail.com) **Based on:** HLD-TFA-Settings v1.0, FRD-TFA-Settings v1.0 **Module:** Settings · **Sub-module:** 06.3 TFA Settings · **Code:** TFAS © 2026 Jojo George. All rights reserved.

Document Control

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version

Related documents: BRD · FRD · HLD · TDD · Specs

Document	Location
BRD – TFA Settings	`../../../BRD/06-Settings/06.3-TFA-Settings/BRD-TFA-Settings.md`
FRD – TFA Settings (FR-TFAS-01…28, UC-TFAS-01…03)	`../../../FRD/06-Settings/06.3-TFA-Settings/FRD-TFA-Settings.md`
HLD – TFA Settings	`../../../HLD/06-Settings/06.3-TFA-Settings/HLD-TFA-Settings.md`
Overall LLD (stack, cross-cutting design, Auth)	`../../00-Overall/LLD-TaskFlow-Overall.md`
LLD – Assistant (chat, `ChatModelGateway` callers, provider selector)	`../../08-TFA-Assistant/08.1-Assistant/LLD-Assistant.md`
LLD – Role Configuration (the four `settings.ai.*` keys and their prerequisite)	`../06.4-Role-Configuration/LLD-Role-Configuration.md`
TDD v2.0	`TDD-TaskFlow.md` §6.5 (integrations), §7.2.6 (`AiProviderConfig`, `AiModel`)

1. Introduction

1.1 Purpose

Detailed design of TFA Settings – the organization-wide configuration of the AI providers behind TFA, the TaskFlow assistant: controller actions, configuration and connection-test services, repositories, key protection, provider clients, the `AiProviderConfigs` and `AiModels` tables, seeding, the Angular screen, tests and FR-TFAS-nn traceability. Everything is taken from the code as built.

1.2 Scope & Conventions

- Covered: `AiProvidersController` (4 actions), `AiProviderConfigService`, `AiProviderConnectionTester`, `AiProviderConfigRepository`, `AiModelRepository`, `AiKeyProtector`,

`KeyRingProtectionCheck`, `AiProviderSeeder`, `AiProviderCatalog`, the provider clients used by the test, and `AiProviderSettingsComponent`.

- Referenced only: `GET /api/v1/assistant/providers` (Assistant sub-module, `assistant.chat`), which returns the configured providers through `AiProviderConfigService.GetAvailableAsync`; chat calls through `IChatModelGateway`.
- Naming: `*Controller`, `I*Service` / `*Service`, `I*Repository` / `*Repository`, DTOs `*Dto` / `*Request`.
- API base path `/api/v1`; JSON camelCase; enums as strings; errors as ProblemDetails.

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2. Code Structure

2.1 Backend Folder Layout

```
TaskFlow.Api/
  Controllers/Settings/AiProvidersController.cs
  Services/Settings/TfaSettings/ProviderConfig/IAiProviderConfigService.cs,
AiProviderConfigService.cs
  Services/Settings/TfaSettings/ConnectionTest/IAiProviderConnectionTester.cs,
AiProviderConnectionTester.cs
  Repositories/Settings/TfaSettings/ProviderConfig/IAiProviderConfigRepository.cs,
AiProviderConfigRepository.cs
  Repositories/Settings/TfaSettings/AiModels/IAiModelRepository.cs, AiModelRepository.cs
  Dtos/Settings/TfaSettings/AiProviderDtos.cs
  Dtos/Settings/TfaSettings/Validators/UpdateAiProvidersRequestValidator.cs   (+
UpdateAiProviderRequestValidator)
  Models/Entities/AiProviderConfig.cs, AiModel.cs
  Models/Enums/AiProviderKind.cs, ProviderFailureKind.cs
  Common/Constants/AiProviderCatalog.cs
  Integrations/Ai/KeyProtection/IAiKeyProtector.cs, AiKeyProtector.cs,
KeyRingProtectionCheck.cs
  Integrations/Ai/Providers/OllamaProvider.cs, GeminiProvider.cs, OpenAiProvider.cs,
ProviderHttp.cs,
                           ProviderFailure.cs, AiProviderRuntime.cs, ChatModelTypes.cs,
IChatModelProvider.cs
  Integrations/Ai/Registry/IChatModelProviderRegistry.cs, ChatModelProviderRegistry.cs
  Services/Assistant/ModelGateway/IChatModelGateway.cs, ChatModelGateway.cs     (Assistant
module)
  Services/Assistant/Chat/ProviderFailureMessages.cs                            (Assistant
module)
  Config/AiProviderOptions.cs, Config/IntegrationRegistration.cs
  Persistence/Seeders/AiProviderSeeder.cs
```

2.2 Frontend Folder Layout

```
src/app/modules/settings/
  settings.routes.ts                                              (route settings/tfa)
  tfa-settings/pages/tfa-settings.component.ts                    (TfaSettingsComponent, page
wrapper)
```

```
  tfa-settings/components/ai-provider-settings.component.{ts,html,css}
  tfa-settings/components/ai-provider-settings.model.ts            (ProviderForm)
  tfa-settings/services/ai-provider-admin.service.ts               (AiProviderAdminService)
src/app/modules/assistant/models/assistant.model.ts                (AiProviderConfig,
UpdateAiProvider, AiProviderTestResult)
```

2.3 Layer Rules

Rule	Enforced by
Controller → Service → Repository only, through interfaces	`ArchitectureTests`
Services commit through `IUnitOfWork`; repositories never save	`ArchitectureTests`
Provider HTTP clients live in `Integrations/`, called only through `IChatModelProviderRegistry` / `IChatModelGateway`	Code structure (`ArchitectureTests` namespace rules)
Child component in `components/`, `loadComponent` target in `pages/`	`npm run check:structure`

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3. Cross-Cutting Design

Shared mechanisms are in the overall LLD. Applied here:

Concern	Design
Validation	`UpdateAiProvidersRequestValidator` (+ per-item `UpdateAiProviderRequestValidator`) in `ValidationFilter`; extra rules in the service throw `RequestValidationException` (400 with `errors`)
Error handling	`GlobalExceptionHandler`; the connection test never fails with an error status
Transactions	One `SaveChangesAsync` per save; on `UniqueConstraintException` duplicate models are detached and saved again
Authorization	Class `[Authorize]` + one `[RequirePermission]` per action (`settings.ai.view`, `settings.ai.edit`, `settings.ai.model.remove`, `settings.ai.test`)
Logging	`ChatModelGateway` logs provider, failure kind and HTTP status only; `GetRuntimeAsync` warns when an enabled key cannot be decrypted; provider HTTP clients have all loggers removed
Secrets	API keys encrypted with ASP.NET Core Data Protection (`AiKeyProtector`), never returned or logged

3.1 Exception → HTTP Status

Exception	Status	When
Validation failure (`ValidationFilter`)	400	Request rules (3 below)
`RequestValidationException`	400	Missing provider field, model length, key for Ollama, remove built-in / in-use model
`ValidationProblem` in the action	400	`DELETE …/models` without `name`
Model binding (`[ApiController]`)	400	`{kind}` not an `AiProviderKind` value
`NotFoundException`	404	Model to remove not in the list

Exception	Status	When
Policy failure	403	Missing the action's permission
`UniqueConstraintException` (not recovered)	409	Unique index broken by something other than a duplicate model
`ProviderNotConfiguredException`, `ProviderFailureException`	200	Caught by `AiProviderConnectionTester` and returned as `ok: false`

4. Module Designs

4.1 Module: Settings

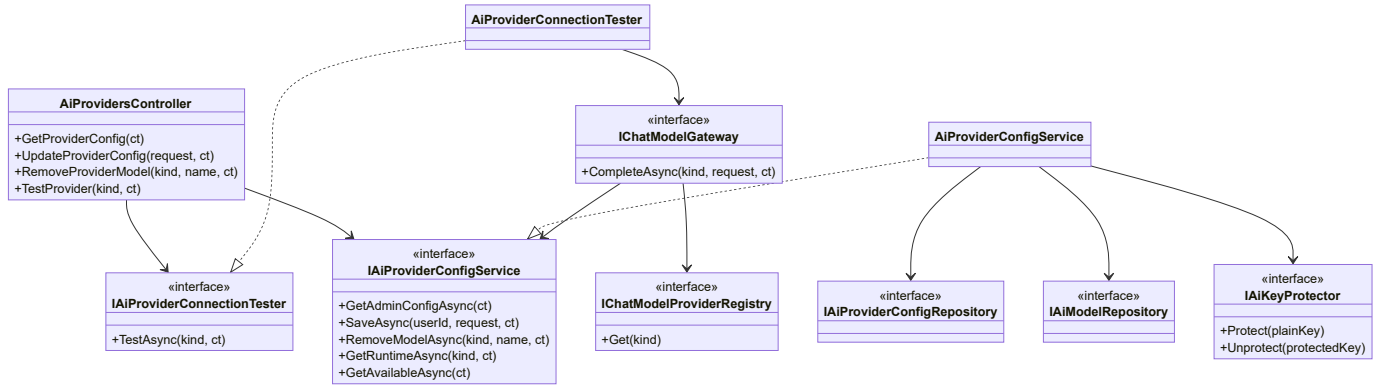
4.1.1 Child module: TFA Settings (TFAS)

Class / file inventory

Layer	Type	File	Responsibility
Controller	`AiProvidersController`	`Controllers/Settings/AiProvidersController.cs`	4 actions under `/api/v1/assistant/providers`
Service	`IAiProviderConfigService` / `AiProviderConfigService`	`Services/Settings/TfaSettings/ProviderConfig/`	Admin view, save, remove model, runtime for calls, available providers
Service	`IAiProviderConnectionTester` / `AiProviderConnectionTester`	`Services/Settings/TfaSettings/ConnectionTest/`	One "Reply with OK." call, result DTO
Repository	`IAiProviderConfigRepository` / `AiProviderConfigRepository`	`Repositories/Settings/TfaSettings/ProviderConfig/`	Provider rows
Repository	`IAiModelRepository` / `AiModelRepository`	`Repositories/Settings/TfaSettings/AiModels/`	Model list, duplicate-race recovery
DTOs	`AiProviderConfigDto`, `AiProviderOptionDto`, `UpdateAiProviderRequest`, `UpdateAiProvidersRequest`, `AiProviderTestResultDto`	`Dtos/Settings/TfaSettings/AiProviderDtos.cs`	Contracts
Validators	`UpdateAiProvidersRequestValidator`, `UpdateAiProviderRequestValidator`	`Dtos/Settings/TfaSettings/Validators/UpdateAiProvidersRequestValidator.cs`	Request rules, SSRF guard `IsMetadataHost`
Entities	`AiProviderConfig`, `AiModel`	`Models/Entities/`	Tables
Enums	`AiProviderKind`, `ProviderFailureKind`	`Models/Enums/`	Values in 5.4
Catalogue	`AiProviderCatalog`	`Common/Constants/AiProviderCatalog.cs`	Display names, locality, key need, default and built-in models, `DefaultOllamaBaseUrl`

Layer	Type	File	Responsibility
Key protection	`IAiKeyProtector` / `AiKeyProtector`, `KeyRingProtectionCheck`	`Integrations/Ai/KeyProtection/`	`Protect` / `Unprotect` (purpose `TaskFlow.AiProviderKeys`); start-up key-ring check
Gateway	`IChatModelGateway` / `ChatModelGateway`	`Services/Assistant/ModelGateway/`	Runtime lookup, provider call, failure logging
Registry / clients	`ChatModelProviderRegistry`, `OllamaProvider`, `GeminiProvider`, `OpenAiProvider`, `ProviderHttp`	`Integrations/Ai/Registry/`, `Integrations/Ai/Providers/`	HTTP calls and failure classification
Messages	`ProviderFailureMessages`	`Services/Assistant/Chat/ProviderFailureMessages.cs`	Fixed user messages per failure kind
Options	`AiProviderOptions`	`Config/AiProviderOptions.cs`	`KeyRingPath`, `TimeoutSeconds`
Seeder	`AiProviderSeeder`	`Persistence/Seeders/AiProviderSeeder.cs`	Built-in models and one row per provider
Frontend page	`TfaSettingsComponent`	`tfa-settings/pages/tfa-settings.component.ts`	Page header + child component
Frontend component	`AiProviderSettingsComponent`	`tfa-settings/components/ai-provider-settings.component.ts`	Forms per provider, linked Model/Server selects, save, test, remove, key status
Frontend service	`AiProviderAdminService`	`tfa-settings/services/ai-provider-admin.service.ts`	`getConfig()`, `save()`, `removeModel()`, `test()`

Class diagram



API contracts

Controller `AiProvidersController`, `[Route("api/v1/assistant/providers")]`, class `[Authorize]`. Endpoints of this sub-module (facts.md §3, `AiProvidersController`): **4**.

#	Method	Route	Permission / policy	Request	Response	Status codes
1	GET	`/api/v1/assistant/providers/config`	`settings.ai.view`	–	`IReadOnlyList<AiProviderConfigDto>` (3 items, catalogue order)	200, 401, 403
2	PUT	`/api/v1/assistant/providers/config`	`settings.ai.edit`	`UpdateAiProvidersRequest`	`IReadOnlyList<AiProviderConfigDto>`	200, 400, 401, 403, 409
3	DELETE	`/api/v1/assistant/providers/{kind}/models`	`settings.ai.model.remove`	route `kind`, query `name`	`IReadOnlyList<AiProviderConfigDto>`	200, 400, 401, 403, 404
4	POST	`/api/v1/assistant/providers/{kind}/test`	`settings.ai.test`	route `kind`, empty body	`AiProviderTestResultDto`	200, 400, 401, 403

`{kind}` binds to `AiProviderKind` (`Ollama`, `Gemini`, `OpenAI`).

DTOs

DTO	Fields
`AiProviderConfigDto`	`Kind`, `DisplayName`, `Locality` (`Local` / `Cloud`), `RequiresKey`, `IsEnabled`, `ModelName`, `BaseUrl` (Ollama only), `KeySet`, `KeyLast4`, `IsConfigured`, `KeyReadable`, `Models` (all names, chosen one added if missing), `CustomModels` (non-built-in)
`UpdateAiProvidersRequest`	`Providers` (`IReadOnlyList<UpdateAiProviderRequest>?`)
`UpdateAiProviderRequest`	`Kind` (`AiProviderKind?`), `IsEnabled` (`bool?`), `ModelName` (`string?`), `BaseUrl` (`string?`), `ApiKey` (`string?`, empty keeps the stored key), `ClearKey` (`bool?`)
`AiProviderTestResultDto`	`Ok`, `Message` (fixed text, never the provider's own error), `FailureKind` (`ProviderFailureKind?`)
`AiProviderOptionDto` (Assistant selector)	`Kind`, `DisplayName`, `ModelName`, `Locality`

Request / response example

```
// PUT /api/v1/assistant/providers/config  (as sent by the screen: only the chosen server enabled)
{ "providers": [
  { "kind": "Ollama", "isEnabled": false, "modelName": "llama3.2:3b", "baseUrl": "http://localhost:11434" },
  { "kind": "Gemini", "isEnabled": true, "modelName": "gemini-2.0-flash", "apiKey": "<secret>" },
  { "kind": "OpenAI", "isEnabled": false, "modelName": "gpt-4o-mini", "clearKey": true }
] }
```

```
// 200 OK (one of three items)
{ "kind": "Gemini", "displayName": "Google Gemini", "locality": "Cloud", "requiresKey": true, "isEnabled": true,
```

```
  "modelName": "gemini-2.0-flash", "baseUrl": null, "keySet": true, "keyLast4": "x9Qa",
"isConfigured": true,
  "keyReadable": true, "models": ["gemini-2.0-flash", "gemini-2.5-flash", "gemini-2.5-pro"],
"customModels": [] }
```

```
// POST /api/v1/assistant/providers/Ollama/test  ->  200 OK
{ "ok": false, "message": "Ollama is not running or cannot be reached. Ask an administrator to
start it, or choose another model.", "failureKind": "Unreachable" }
```

```
// DELETE /api/v1/assistant/providers/Ollama/models?name=llama3.2:3b  ->  400
{ "status": 400, "errors": { "name": ["A built-in model cannot be removed."] } }
```

Service logic – `AiProviderConfigService.GetAdminConfigAsync`

1. `_configs.GetAllAsync()` (no tracking) and `LoadModelsAsync()` → `ModelLists(All, Custom)` lookups by `Kind` (built-in first, then by `Id`).
2. For each `AiProviderCatalog.All` kind (`Ollama`, `Gemini`, `OpenAI`) `ToDto(kind, row, models)`: chosen model = row's `ModelName` or `AiProviderCatalog.DefaultModel(kind)`, added to `Models` if missing; `BaseUrl` = row value or `DefaultOllamaBaseUrl` for Ollama, null for cloud; `KeySet` = encrypted key present; `KeyReadable` = no key, or `Unprotect` succeeds; `IsConfigured` = `IsConfigured(row)`.
3. `IsConfigured(row)`: enabled and model not blank, and either (no key needed and `BaseUrl` not blank) or (encrypted key present and decryptable) (FR-TFAS-16, FR-TFAS-22).

Service logic – `AiProviderConfigService.SaveAsync(userId, request)`

1. `now`; `rows = _configs.GetAllForUpdateAsync()` (tracked list).
2. For each update: null `Kind`, `IsEnabled` or `ModelName` → `RequestValidationException("providers", "Every provider setting must be provided.")`.
3. `modelName = ModelName.Trim()`; length 0 or over 100 → "The `DisplayName` model name must be 1 to 100 characters."
4. Find or create the row (`new AiProviderConfig { Kind }`, `Add`). Set `IsEnabled`, `ModelName`.
5. When enabled, `RememberModelAsync`: if `IsKnownAsync(kind, name)` is false (checks `Set.Local` and the database), add `AiModel { IsBuiltIn = false }` (FR-TFAS-05).
6. `BaseUrl`: Ollama → `NormalizeBaseUrl` (trim, trailing `/` removed) or `DefaultOllamaBaseUrl`; cloud → null.
7. Key: non-empty `ApiKey` on a provider that needs no key → "Ollama does not use an API key." (FR-TFAS-09); otherwise `EncryptedApiKey = _keyProtector.Protect(key.Trim())`, `KeyLast4` = last 4 characters. Else `ClearKey == true` → both null (FR-TFAS-13). Empty key and no clear → unchanged (FR-TFAS-12).
8. `UpdatedByUserId`, `UpdatedAt`. `SaveKeepingAnyDuplicateModelAsync`: on `UniqueConstraintException`, `DropAddedModelsStoredElsewhereAsync()` detaches staged duplicates and saves again; otherwise rethrows.
9. Reload models and return the three DTOs. The backend accepts several enabled providers; the screen enables only the selected one (FR-TFAS-14).

Service logic – `AiProviderConfigService.RemoveModelAsync(kind, name)`

1. `GetForUpdateAsync(kind, name)` → null → `NotFoundException("That model is not in the list.")`.
2. `IsBuiltIn` → `RequestValidationException("name", "A built-in model cannot be removed.")`.
3. `UsesModelAsync(kind, name)` → "That model is the one saved for its server. Choose another model and save before removing it."

4. `Remove`, save, return `GetAdminConfigAsync()` (FR-TFAS-20, FR-TFAS-21).

Service logic – `GetRuntimeAsync(kind)` and `AiProviderConnectionTester.TestAsync(kind)`

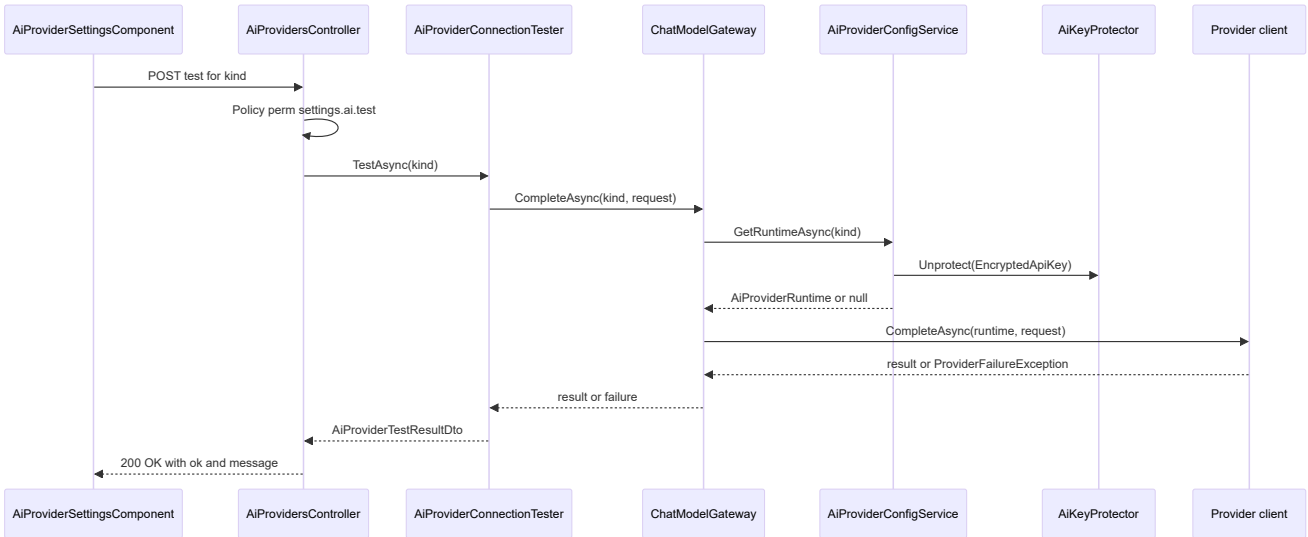
1. `GetRuntimeAsync`: row missing or not configured → null (warning logged when the row is enabled and its key cannot be decrypted); else `AiProviderRuntime(kind, ModelName, BaseUrl, decryptedKey)` – the key lives only in memory.
2. `TestAsync` builds `ChatModelRequest` with system text "This is a connection test. Reply with the single word OK.", one user message "Reply with OK.", no tools.
3. `IChatModelGateway.CompleteAsync(kind, request)`: runtime null → `ProviderNotConfiguredException`; else `IChatModelProviderRegistry.Get(kind).CompleteAsync(runtime, request)`.
4. Success → `(true, "The connection works.", null)`; `ProviderNotConfiguredException` → `ProviderFailureMessages.NotConfigured(kind)`; `ProviderFailureException` → `ProviderFailureMessages.For(kind, failure.Kind, failure.RetryAfter)` and `failure.Kind` (FR-TFAS-17, FR-TFAS-18).

Provider clients (used by the test)

Client	Call	Key
`OllamaProvider`	`POST {BaseUrl}/api/chat`	none
`GeminiProvider`	`POST https://generativelanguage.googleapis.com/v1beta/models/{model}:generateContent`	header `x-goog-api-key`
`OpenAiProvider`	`POST https://api.openai.com/v1/chat/completions`	`Authorization: Bearer`

`ProviderHttp.SendAsync` maps a client timeout to `Timeout` and `HttpRequestException` to `Unreachable`; unparsable bodies to `Unexpected`; status and provider error codes map to `RateLimited`, `InvalidKey`, `ModelUnavailable`. HTTP clients: `AllowAutoRedirect = false`, `RemoveAllLoggers()`, timeout `AiProviders:TimeoutSeconds`.

Sequence diagram – connection test



Sequence diagram – save



Repository methods

Method	Query / purpose	Notes
`AiProviderConfigRepository.GetAllAsync()`	All rows	`AsNoTracking`
`AiProviderConfigRepository.GetAllForUpdateAsync()`	All rows, tracked `List`	Save
`AiProviderConfigRepository.GetAsync(kind)`	One row	`AsNoTracking`; `IX_AiProviderConfigs_Kind`
`AiProviderConfigRepository.UsesModelAsync(kind, modelName)`	Is it the saved model	`AnyAsync`
`AiModelRepository.GetAllAsync()`	All models	`AsNoTracking`, `OrderByDescending(IsBuiltIn).ThenBy(Id)`
`AiModelRepository.IsKnownAsync(kind, name)`	Stored or staged in this request	`Set.Local` then `AnyAsync`
`AiModelRepository.GetForUpdateAsync(kind, name)`	Tracked model	`FirstOrDefaultAsync`; `IX_AiModels_Kind_Name`
`AiModelRepository.DropAddedModelsStoredElsewhereAsync()`	Race recovery	Detaches `Added` entries already stored; returns whether any was dropped
`Add` / `Remove` (base)	Stage	`BaseRepository`

Validation rules

Field	Rule	Message
`Providers`	`NotNull()`	"Providers are required."
`Providers`	Distinct `Kind`	"Each provider can only appear once."

Field	Rule	Message
`Kind`	`NotNull()`, `IsInEnum()`	FluentValidation default
`IsEnabled`	`NotNull()`	FluentValidation default
`ModelName`	`NotEmpty()`, `MaximumLength(100)`	"Model name cannot be blank."
`BaseUrl`	blank, or absolute `http`/`https` URI not matching `IsMetadataHost` (`metadata.google.internal`, `metadata`, `instance-data`, 0.0.0.0, `::`, IPv6 link-local, 169.254.x.x)	"The server address must be a valid http or https URL, and not a link-local or cloud metadata address."
`BaseUrl`	`MaximumLength(300)`	FluentValidation default
`ApiKey`	`MaximumLength(500)`, no whitespace	"The API key cannot contain spaces."
item	not (`ClearKey == true` and non-empty `ApiKey`)	"Provide a new key or clear the key, not both."
Service	missing field / model length / key for Ollama	see service logic
Controller	`name` blank on DELETE	"The model name is required."

Client mirror: `modelName` required, max 100; `baseUrl` pattern `^(https?:\/\/\S+)?$`; `apiKey` pattern `^\S*$`, max 500; group validator `newModelRequired` (1–100 characters when **Add new model…** (`ADD_MODEL`) is chosen).

Error cases

Case	Exception / result	Status
Missing the action's `settings.ai.*` permission	Policy refusal (logged)	403
Invalid request	`ValidationFilter` / `RequestValidationException`	400
Unknown `kind` in the route	Model binding	400
Model not in the list	`NotFoundException`	404
Built-in or saved model	`RequestValidationException`	400
Provider not configured / unreachable / key refused	Test result `ok: false`	200

Frontend design – `AiProviderSettingsComponent`

Member	Behaviour
`visible`, `canEdit`, `canTest`, `mayRemoveModel`	`AuthService.hasPermission` for `SettingsAiView`, `SettingsAiEdit`, `SettingsAiTest`, `SettingsAiRemoveModel`; Save, Test and Remove buttons rendered only when allowed
`load()` / `apply()`	`getConfig()`; active kind = first enabled and configured (else first enabled); one `ProviderForm` per provider; key boxes start empty

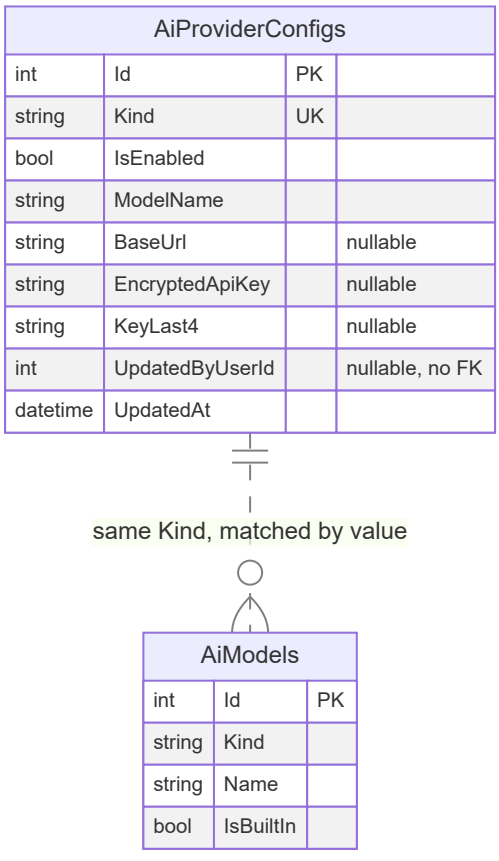
Member	Behaviour
`chooseModel()` / `selectKind()`	Linked Model and Server selects (value `kind
`statusText()`	"In use" / "Not configured" for the active kind; otherwise "Needs an API key" or "Ready to use" (FR-TFAS-02)
`keyStatus()`	"No key saved yet.", "The saved key can no longer be read. Enter it again.", "A key is saved (ends … `last4`). Leave the box empty to keep it.", "The stored key will be removed when you save."
`clearKey()` / `keepKey()`	Marks a key for removal / undoes it
`save()`	Validates all forms (shows the first invalid provider); sends all three providers with `isEnabled` true only for the selected one; toast "TFA settings saved"; error "Could not save the TFA settings."
`test()`	If `needsSave()`, saves first then runs the test; result per kind shown next to the button; HTTP error → "The test could not be run."
`removeModel()`	Only for a custom, not-saved model; reloads from the response; toast "Removed `name` from the list"; error "Could not remove the model."

State machine: *Not applicable for this sub-module.* (A provider is only enabled or disabled; the screen status is computed, not stored.)

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5. Database Design

5.1 ER Diagram



5.2 Tables

`AiProviderConfigs` (AiProviderConfig)

Column	Type	Null	Key / index	Default	Description
Id	int (INTEGER)	No	PK	identity	Row id
Kind	`AiProviderKind` as string(16) (TEXT)	No	Unique `IX_AiProviderConfigs_Kind`	–	`Ollama`, `Gemini`, `OpenAI`
IsEnabled	bool (INTEGER)	No	–	seeded (Ollama on for a fresh install)	Provider switched on
ModelName	string(100) (TEXT)	No	–	`AiProviderCatalog.DefaultModel`	Chosen model
BaseUrl	string(300) (TEXT)	Yes	–	`http://localhost:11434` for Ollama	Ollama only; null for cloud
EncryptedApiKey	string (TEXT)	Yes	–	null	Data Protection ciphertext; never returned
KeyLast4	string(4) (TEXT)	Yes	–	null	Display hint
UpdatedByUserId	int? (INTEGER)	Yes	– (no FK)	null	Last editor
UpdatedAt	DateTime (TEXT)	No	–	–	UTC

`AiModels` (AiModel)

Column	Type	Null	Key / index	Default	Description
Id	int (INTEGER)	No	PK	identity	Row id; orders added models
Kind	`AiProviderKind` as string(16) (TEXT)	No	Unique `IX_AiModels_Kind_Name` (with `Name`)	–	Provider
Name	string(100) (TEXT)	No	Unique with `Kind`	–	Model name
IsBuiltIn	bool (INTEGER)	No	–	–	True = from `AiProviderCatalog.BuiltInModels`; false = added by an administrator

Built-in models (seeded): Ollama `llama3.2:3b`, `llama3.2:1b`, `llama3.1:8b`, `qwen2.5:7b`, `mistral:7b`; Gemini `gemini-2.0-flash`, `gemini-2.5-flash`, `gemini-2.5-pro`; OpenAI `gpt-4o-mini`, `gpt-4o`, `gpt-4.1-mini` (first = default).

5.3 Indexes, Constraints & Triggers

Object	Definition	Purpose
`IX_AiProviderConfigs_Kind`	Unique (`Kind`)	One row per provider

Object	Definition	Purpose
`IX_AiModels_Kind_Name`	Unique (`Kind` , `Name`)	No duplicate model; the save recovers from a duplicate race
–	No FK between the tables; no triggers	Matched by `Kind` value

5.4 Enumerations

Enum	Values	Stored as
`AiProviderKind`	`Ollama` , `Gemini` , `OpenAI`	Text (`HasConversion<string>()` , max 16); JSON string; route value
`ProviderFailureKind`	`RateLimited` , `InvalidKey` , `Timeout` , `Unreachable` , `ModelUnavailable` , `Unexpected`	Not stored; JSON string in `AiProviderTestResultDto.FailureKind`

5.5 Migrations

Migration	Change
`20261018183210_AddAssistant`	Creates `AiProviderConfigs` and `IX_AiProviderConfigs_Kind` (with the assistant tables)
`20261019163241_AddAiModels`	Creates `AiModels` and `IX_AiModels_Kind_Name`

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6. Algorithms & Background Jobs

Job / algorithm	Trigger / schedule	Steps	Idempotency
`AiProviderSeeder.SeedAsync`	Every start-up, after `AccessControlSeeder` , before `DatabaseSeeder`	1. Adds missing built-in models. 2. Adds each saved `ModelName` not yet listed as a non-built-in model. 3. Adds a missing provider row: model = default, Ollama `BaseUrl` = default, `IsEnabled` = Ollama and no user exists yet (fresh install) (FR-TFAS-24)	Only inserts what is missing
`KeyRingProtectionCheck.Check`	Start-up on Windows (`Program.cs`)	Counts `key-*.xml` files with a plain `masterKey` ; logs a warning with the fix, or an information line when it could not check	Read-only, never throws
Duplicate-model recovery	On `UniqueConstraintException` during save	Detach staged duplicates, save again	Keeps the other request's row

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7. Security Design (detailed)

Item	Implementation
Permission checks	`settings.ai.view` (GET), `settings.ai.edit` (PUT), `settings.ai.model.remove` (DELETE), `settings.ai.test` (POST); catalogue `Requires: settings.ai.view` for the other three (FR-TFAS-25, FR-TFAS-26)

Item	Implementation
Defaults	Administrator only on first seed (Manager and Employee not granted); legacy `settings.ai.manage` expands to the four keys in `EffectivePermissions.LegacyChildren`
Key storage	`AiKeyProtector` (Data Protection purpose `TaskFlow.AiProviderKeys`, application name `TaskFlow`); key ring at `Storage:Root` + `AiProviders:KeyRingPath`; `ProtectKeysWithDpapi()` on Windows
Key exposure	Never returned (only `KeySet`, `KeyLast4`, `KeyReadable`); decrypted only in `AiProviderRuntime` for one call; not logged (FR-TFAS-10, FR-TFAS-11)
SSRF	Validator refuses link-local and cloud-metadata hosts; HTTP clients do not follow redirects; cloud addresses are fixed in code
Logging	Provider clients log nothing; gateway logs provider, failure kind and HTTP status only
Frontend	`permissionGuard` with `SettingsAiView` on `settings/tfa`; menu item "TFA Settings"; in-component `canEdit`, `canTest`, `mayRemoveModel` (FR-TFAS-27)
Cloud notice	Screen text explains that Gemini and OpenAI are cloud services (FR-TFAS-23)

8. Configuration

Key	Default	Purpose
`AiProviders:KeyRingPath`	`dp-keys`	Data Protection key-ring folder, relative to `Storage:Root`
`AiProviders:TimeoutSeconds`	`60`	Timeout of one provider HTTP call
`Storage:Root`	`""`	Root folder for the key ring and database
API keys	`<secret>`	Entered on the screen, stored encrypted in `AiProviderConfigs.EncryptedApiKey`; never in `appsettings.json`

9. Testing Design

"Covers" was checked against the test sources. Backend paths are relative to `backend/tests`; frontend paths to `frontend/src/app/modules/settings/tfa-settings`.

Area	Test class / file	Type	Covers
Service	`TaskFlow.UnitTests/Settings/TfaSettings/ProviderConfig/AiProviderConfigServiceTests.cs`	Unit	Not-configured states, model lists and custom models, model name rules, concurrent model add, key last four / encrypted / kept / replaced / cleared, unreadable key reported and logged without key material, validator
Seeder	`TaskFlow.UnitTests/Settings/TfaSettings/ProviderConfig/AiProviderSeederTests.cs`	Unit	Built-in models, Ollama on and cloud providers off on a fresh database, all off on an upgrade, idempotent, saved values never overwritten
Key protection	`TaskFlow.UnitTests/Integrations/Ai/KeyProtection/AiKeyProtectorTests.cs`, `KeyRingProtectionCheckTests.cs`	Unit	Protect / unprotect round trip, purpose isolation, replaced key ring gives null; counting of unprotected key-ring files

Area	Test class / file	Type	Covers
Provider clients	`TaskFlow.UnitTests/Integrations/Ai/Providers/ChatProviderTests.cs`	Unit	Ollama, Gemini and OpenAI requests, tool calls and failure classification; messages never contain the key
Gateway	`TaskFlow.UnitTests/Assistant/ModelGateway/ProviderFailureHandlingTests.cs`	Unit	Failure messages and retry time, no cross-provider fallback, per-user rate limit
Endpoints	`TaskFlow.IntegrationTests/Settings/TfaSettings/AssistantProviderEndpointsTests.cs`	Integration	Non-administrators 403, 401, save without the key in any response or log, DPAPI key ring on Windows, invalid input 400, test-connection results, remove an added model only
First run	`TaskFlow.IntegrationTests/Settings/TfaSettings/AiDefaultsOnFirstRunTests.cs`	Integration	Fresh database has the model list and Ollama ready; every signed-in user can use TFA on the first run
SSRF	`TaskFlow.IntegrationTests/Integrations/Ai/ProviderRedirectTests.cs`	Integration	Redirects not followed
Start-up warning	`TaskFlow.IntegrationTests/Common/KeyRingStartupWarningTests.cs`	Integration	Key-ring warning at start-up
Support	`TaskFlow.IntegrationTests/TestSupport/ScriptedChatProvider.cs`, `CapturingLoggerProvider.cs`	Support	Fake provider, captured logs

| Frontend panel | `components/ai-provider-settings.component.spec.ts` | Frontend unit | Hidden from non-administrators, server dropdown and model lists, add a model, one server in use, key never shown / kept / replaced / removed, invalid input, save error, test connection (saves first, failure messages), removing an added model, load retry | | Frontend page | `pages/tfa-settings.component.spec.ts` | Frontend unit | Own screen and title; nothing shown or requested without `settings.ai.manage` |

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10. Known Limitations & Open Issues

- The backend accepts several enabled providers; only the screen enforces one active server.
- Without `settings.ai.edit` the inputs stay editable; only **Save** is hidden.
- `test()` saves unsaved changes first; a user with `settings.ai.test` but not `settings.ai.edit` gets 403 from that save and the test does not run.
- `GetAdminConfigAsync` calls `Unprotect` up to twice per provider per request (`KeyReadable`, `IsConfigured`).
- A lost or replaced key ring makes saved keys unreadable; they must be entered again (shown on the screen).
- > ⚠️ To be confirmed (FRD open issues): whether view-only users should get a fully read-only screen, whether a test-only user should test the saved configuration, whether the server should refuse more than one active provider, and the exact rule for **Ready to use** versus **In use**.

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11. Traceability (FRD → code)

FRD requirement	Endpoint / class	Test
FR-TFAS-01	`AiProviderKind`, `AiProviderCatalog` (`RequiresKey`, `Locality`)	`AiProviderConfigServiceTests`
FR-TFAS-02	`AiProviderConfigDto` · `statusText()` · `activeKind`	`AssistantProviderEndpointsTests`
FR-TFAS-03	`AiProviderCatalog.BuiltInModels` · `AiProviderSeeder` · `AiModelRepository.GetAllAsync`	`AiProviderSeederTests`, `AiDefaultsOnFirstRunTests`
FR-TFAS-04	`chooseModel()`, `selectKind()`	`ai-provider-settings.component.spec.ts`
FR-TFAS-05	`ADD_MODEL`, `newModelRequired` · `RememberModelAsync`	`AiProviderConfigServiceTests`
FR-TFAS-06	`PUT /api/v1/assistant/providers/config` · `NormalizeBaseUrl`	`AiProviderConfigServiceTests`
FR-TFAS-07	`UpdateAiProviderRequestValidator.BeHttpUrl` / `IsMetadataHost` · `AllowAutoRedirect = false`	`AssistantProviderEndpointsTests`, `ProviderRedirectTests`
FR-TFAS-08	Validator `ApiKey` rules	`AssistantProviderEndpointsTests`
FR-TFAS-09	`SaveAsync` "Ollama does not use an API key."	`AiProviderConfigServiceTests`
FR-TFAS-10	`AiKeyProtector.Protect` · DTO without key	`AiKeyProtectorTests`, `AssistantProviderEndpointsTests`
FR-TFAS-11	`KeySet`, `KeyLast4` · `keyStatus()`	`AiProviderConfigServiceTests`
FR-TFAS-12	`SaveAsync` (empty key keeps stored key)	`AiProviderConfigServiceTests`
FR-TFAS-13	`ClearKey` · validator "Provide a new key or clear the key, not both." · `clearKey()` / `keepKey()`	`AiProviderConfigServiceTests`
FR-TFAS-14	`save()` sends `isEnabled` only for the selected kind · toast	`ai-provider-settings.component.spec.ts`
FR-TFAS-15	Validator "Each provider can only appear once."	`AssistantProviderEndpointsTests`
FR-TFAS-16	`IsConfigured` · `GetAvailableAsync` (Assistant `GET /api/v1/assistant/providers`)	`AiProviderConfigServiceTests`
FR-TFAS-17	`POST /api/v1/assistant/providers/{kind}/test` · `AiProviderConnectionTester.TestAsync` · `ProviderFailureMessages`	`AssistantProviderEndpointsTests`, `ProviderFailureHandlingTests`
FR-TFAS-18	Tester catches `ProviderNotConfiguredException` / `ProviderFailureException` → 200	`AssistantProviderEndpointsTests`

FRD requirement	Endpoint / class	Test
FR-TFAS-19	`test()` → `save(() => runTest(kind))` when `needsSave()`	`ai-provider-settings.component.spec.ts`
FR-TFAS-20	`DELETE /api/v1/assistant/providers/{kind}/models` · `RemoveModelAsync` · `removeModel()`	`AssistantProviderEndpointsTests`
FR-TFAS-21	`RemoveModelAsync` (404, built-in, in use)	`AiProviderConfigServiceTests`
FR-TFAS-22	`KeyReadable` · `IsConfigured` · `GetRuntimeAsync` warning	`AiProviderConfigServiceTests`, `AiKeyProtectorTests`
FR-TFAS-23	Cloud notice in `ai-provider-settings.component.html`	`ai-provider-settings.component.spec.ts`
FR-TFAS-24	`AiProviderSeeder.SeedProvidersAsync` (`freshInstall`)	`AiProviderSeederTests`, `AiDefaultsOnFirstRunTests`
FR-TFAS-25	`[RequirePermission]` per action · `canEdit` / `canTest` / `mayRemoveModel`	`AssistantProviderEndpointsTests`, `PermissionAuthorizationTests`
FR-TFAS-26	`PermissionCatalog` `Requires: settings.ai.view` · `RoleService.ResolvePermissionsAsync`	`PermissionCatalogTests`
FR-TFAS-27	`GET /api/v1/assistant/providers/config` · `permissionGuard` (`settings/tfa`) · `buildNavigation()`	`AssistantProviderEndpointsTests`
FR-TFAS-28	Toasts and error texts in `AiProviderSettingsComponent`	`ai-provider-settings.component.spec.ts`

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